

Baochen Tian

Research Assistant, Australian National University

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EDUCATION

Australian National University

Graduated 2025

Bachelor of Science in Physics | GPA: 6.13 / 7.0

Attended 2023 - 2025

Exchange: National Taiwan University, Spring 2025

Beijing Institute of Technology

2020 - 2023

Undergraduate studies; transferred to ANU in 2023

International Economics and Trade (2020 - 2021); Optical-Electronics Information Science (2021 - 2023)

INDEPENDENT PROJECTS

Flow Matching Reimplementation of DiffPhyCon

2026

Derived and implemented a flow-matching formulation of DiffPhyCon's joint trajectory model, matching the original model capacity and training configuration. Compared reconstruction and control against DDPM/DDIM baselines on 1D Burgers and 2D Jellyfish.

Investigated guidance and prior reweighting under flow matching, including behavior unexplained by a naive score-to-velocity conversion and the need for an FM-specific derivation.

Code: github.com/bcTiann/diffphycon

RESEARCH EXPERIENCE

Python Pipeline for Post-processing QUOKKA Simulations

April 2026 - Present

Research School of Astronomy and Astrophysics, ANU.

Supervisor: Dr. ChongChong He.

Investigating how reliably emission tracers (CO, H-alpha, C+, H I) recover the mass-weighted velocity dispersion in simulations, and how this depends on gas phase.

Developing a Python pipeline that converts 3D QUOKKA snapshots into synthetic emission-line observations with DESPOTIC, Grackle and Cloudy.

Code: github.com/bcTiann/quokka_postprocess_3D

Deep Learning for CMB Inpainting

February - June 2025

LeCosPA, National Taiwan University, during an exchange semester.

Supervisor: Prof. Pisin Chen; mentor: Reyhan D. Lambaga.

Applied Transformers and DDPMs to inpaint CMB maps masked by the galactic plane. Evaluated reconstruction fidelity by comparing the angular power spectra of generated maps against ground truth.

Rectangular Haloscope for Axion Detection

January - February 2024

Quantum Technologies and Dark Matter Research Lab, University of Western Australia.

Supervisors: Prof. Michael Tobar and Dr. Aaron Quiskamp.

Modelled a tunable-wall rectangular haloscope in COMSOL, simulating resonant modes and evaluating sensitivity against cylindrical tuning-rod geometries.

SCHOLARSHIPS & SKILLS

EQUUS Summer Research Scholarship

January - February 2024

Six-week research program at the University of Western Australia, supported by the ARC Centre of Excellence for Engineered Quantum Systems.

Machine learning: PyTorch, Transformers, VAEs, diffusion models and flow matching.

Astrophysical modelling: QUOKKA post-processing, synthetic emission-line observations, DESPOTIC, Grackle and Cloudy.

OTHER INTERESTS

Chamber music and performance: Founded chamber ensembles at BIT and ANU; pianist for the BIT University Choir; participated in two university music competitions and 7+ community performances. Performed in NTU's 2025 Symphony Orchestra Chamber Music Night and was invited to perform at an official ANU welcome dinner for visiting scholars.